

Sanjana Kiran Urs

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PROFESSIONAL SUMMARY

ML focused Software Engineer with a graduate degree in Computer Science and hands-on expertise spanning end-to-end ML workflows: from ETL & statistical modeling to NLP and cloud-based inference. Passionate about applying strong foundations in statistics, Python, and SQL to solve real-world problems. Driven by mentorship and a track record of translating analytical insights into production systems that balance model performance with operational efficiency.

EXPERIENCE

Visiting Graduate Student

Los Angeles, California, USA

Cedars Sinai Medical Center - Slomka Lab

July 2025 – August 2025

- Developed an interpretable **XGBoost** model to detect coronary artery disease in **3 weeks** using PET-derived imaging biomarkers and clinical variables from 11 multicenter cohorts. Implemented **LOSO cross-validation** (train on 10 sites, test on 1) to ensure cross-site generalizability, improving **AUC from 0.81 to 0.84**.
- Coordinated integration and standardization of cardiac PET/CT imaging biomarkers across multiple sites, built quality-control pipelines and harmonization protocols, **which reduced data ingestion time and ensured consistent analysis**.
- Accepted for Poster Presentation at American College of Cardiology (ACC)**

PROJECTS

Faculty Portal | *MySQL, Java Spring Boot + React/Next.js, Project Management*

- Architected and delivered a full-stack faculty portal end-to-end, serving **7+ REST APIs across 6 domains**. Worked on improving data consistency, schema structure, and API performance.
- Redesigned a **14-table normalized schema** - eliminating redundant joins and cutting **average query response time by 40% across 8+ workflows** while **reducing peak DB read load by 35%**.

Emotion Detection in Social Media Posts | *Python, PyTorch, HuggingFace, Scikit-learn, Gradio*

- Fine-tuned and compared three Transformer architectures (**RoBERTa, RoBERTa+CNN+NRC-lexicon, DeBERTa**) on Google's GoEmotions dataset with WordPiece tokenization and class weighting - achieving **85% test accuracy** and **reduced test loss from 0.68 to 0.11** with the enhanced model, with **DeBERTa yielding the highest F1 and ROC-AUC scores**.

University Shop Management System | *PostgreSQL, Indexing, Triggers, Python, NumPy, Pandas, Tableau*

- Designed a normalized PostgreSQL DB, implemented triggers, stored procedures, and complex queries to enforce data integrity and surface insights like top-rated shops and highest-spending users.
- Optimized query performance through strategic indexing and sub-query refactoring, achieving up to **10× reduction in execution cost** and enabling real-time Tableau dashboards for revenue and loyalty reporting.

Cricket Performance Analytics using Tableau | *Tableau, Python, NumPy, Pandas* — Article

- Developed interactive dashboards to visualize and analyze cricket performance data across ODI, T20, and Test formats. Visualized player statistics, top performers, and team comparisons using bar charts, pie charts, and bubble charts for insightful analysis.

SKILLS

AI: RAG, GenAI, Transformers, LLM fine tuning, Llama, Langchain, Agentic workflows, Prompt Engineering

Libraries: NumPy, Pandas, SkLearn, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Huggingface

Databases & Big Data: MySQL, PostgreSQL, Spark, Hadoop, Tableau, PowerBI, Excel

Systems: Data Structures, Algorithms, System Design, SDLC, CI/CD

Tools and Frameworks: JIRA, Postman, Linux, Docker, Git

Data Science: Mathematics, Regression, ETL, Advance analytics, Statistics, Data modelling, AWS

Languages: Python, C, HTML, CSS, SQL, R, OOP in C++

Soft Skills: Problem-solving, Communication, High collaboration, Critical thinking, Decision making, CRM

EDUCATION

Master's degree in Computer Science

New York, USA

University at Buffalo - SUNY | GPA: (3.7/4.0)

August 2024 - December 2025

Coursework: Algorithms, Machine Learning, Operating Systems, Deep Learning, Data Models & Query Language , Statistical Data Mining, Data Intensive Computing, Computer Vision, Intro to Pattern Recognition.

Bachelor's degree in Information Science

Mysore, India

JSS Science and Technology University | GPA: (3.87/4.0)

August 2019 - July 2023

Coursework: Artificial Intelligence, Data Mining, Database Management System, Object Oriented Programming in C++, Software Engineering, Linear Algebra, Data Structures & Algorithms, Operating Systems.